Customer Segmentation and Predictive Modelling

END-TO-END DATA SCIENCE PROJECT SUMMARY & JOURNEY

1. Problem Statement

In the modern insurance and financial services landscape, understanding customer behavior and accurately predicting future financial commitments are paramount for sustainable growth. Insurance organizations accumulate vast amounts of customer demographic, behavioral, and policy data, yet frequently translate these disparate records into actionable operational insights. Without precise customer segmentation, marketing strategies remain generalized, leading to suboptimal campaign ROI, increased customer churn, and missed cross-selling opportunities. Furthermore, the inability to reliably predict financial metrics such as premium commitments hinder proactive risk assessment and personalized product pricing.

This project addresses this challenge by establishing an end-to-end data science pipeline. The primary objectives are twofold: (1) to segment customers into distinct, behaviorally cohesive groups using unsupervised machine learning based on demographics, financial metrics, and tenure; and (2) to build a robust predictive regression model capable of accurately forecasting future purchasing patterns and premium commitments from historical attributes.

2. The Project Journey & Story

Chapter 1: The Foundation (Data Cleaning)

We loaded our raw dataset containing 53,503 records into Python using Pandas. Rigorous audits uncovered 13,244 redundant records linked to repeat Customer IDs. We cleaned these up by filtering for unique customer entries, standardizing date-time formats, and verifying numerical types, establishing a pristine working foundation of 40,259 unique profiles.

Chapter 2: Crafting Signals (Feature Engineering)

To extract predictive power, we engineered customer tenure days from purchase history, created a financial efficiency 'Premium-to-Coverage Ratio', binned ages into generational brackets, and applied one-hot encoding across categorical variables, expanding our feature space to 33 rich dimensions.

Chapter 3: Discovering Hidden Tribes (Customer Segmentation)

Using StandardScaler normalization and the Elbow Method across Within-Cluster Sum of Squares (WCSS), we determined an optimal cluster count of $k=4$. K-Means clustering successfully partitioned our customer base into four distinct behavioral tribes.

Chapter 4: Gazing into the Future (Predictive Modelling & Evaluation)

We trained a Random Forest Regressor on an 80/20 train-test split (17,101 fully populated rows) to forecast customer premium amounts. The model achieved an exceptional RMSE of 31.98 and an R-squared score of 0.9994, explaining 99.94% of premium variance.

Chapter 5: Strategic Action (Insights & Recommendations)

We translated technical success into value by designing VIP retention tiers for high-value core clients, digital onboarding flows for young aspirants, and anniversary bundled promotions for mid-tier stability segments.

3. Summary of Workflow & Tasks

Project Task	Core Actions & Technical Implementation
Task 1: Data Cleaning	Loaded raw CSV, audited nulls, removed 13,244 duplicate IDs, formatted datetime fields, and verified numerical data types.
Task 2: Feature Engineering	Computed customer tenure days, engineered premium-to-coverage ratio, binned continuous age, and performed one-hot encoding.
Task 3: Segmentation	Applied StandardScaler normalization, utilized the Elbow Method, and executed K-Means clustering ($k=4$).
Task 4 & 5: Modelling & Evaluation	Trained a Random Forest Regressor (80/20 split) and evaluated model performance, achieving an RMSE of 31.98 and an R^2 of 0.9994.
Task 6: Insights & Export	Synthesized tailored segment strategies, built targeted marketing recommendations, and exported the finalized dataset to CSV.